Predicting SO2 in a city we have never seen

STTN327 project | 41 US cities, 1969–1971 | four models, one fair test

The question we set out to answer

Predict SO2 from weather, season, city size and industry, including for a city that is not in our data.

Testable aim: a model succeeds if its error for an unseen city is clearly below the benchmark of guessing the average (RMSE 23.98).

1

RQ1: What drives SO2 in the 41 cities?

2

RQ2: How well can SO2 be predicted for a new city, with no, one or two quarters of its readings?

3

RQ3: Which of four models predicts best?

Why four models instead of one?

Regression

Simple, interpretable baseline

What does the plain linear story say?

Ridge / Lasso

Shrinks coefficients; copes with correlated predictors (r = 0.96)

Does overfitting or collinearity hurt?

Random forest

No assumed shape; finds curves and interactions

Are we missing non-linear effects?

Mixed-effects

Gives every city its own baseline

Does the city structure matter?

Same test for all four. If different approaches agree, the answer is not an accident of one model; where they disagree, we learn why.

Roadmap

1. Data and its problems
2. One fair test for all models
3. The four models, one at a time
4. Head-to-head comparison
5. Conclusion and limits

The data: from monthly Excel to 492 quarterly rows

1

Fill gaps

Missing months interpolated within each city

2

Group

36 months → 12 quarters (Q1 = Jan–Mar …)

3

Summarise

Average SO2, temperature, wind, precipitation; sum precipitation days

4

Join

Add manufacturing count and population per city

41

US cities

492

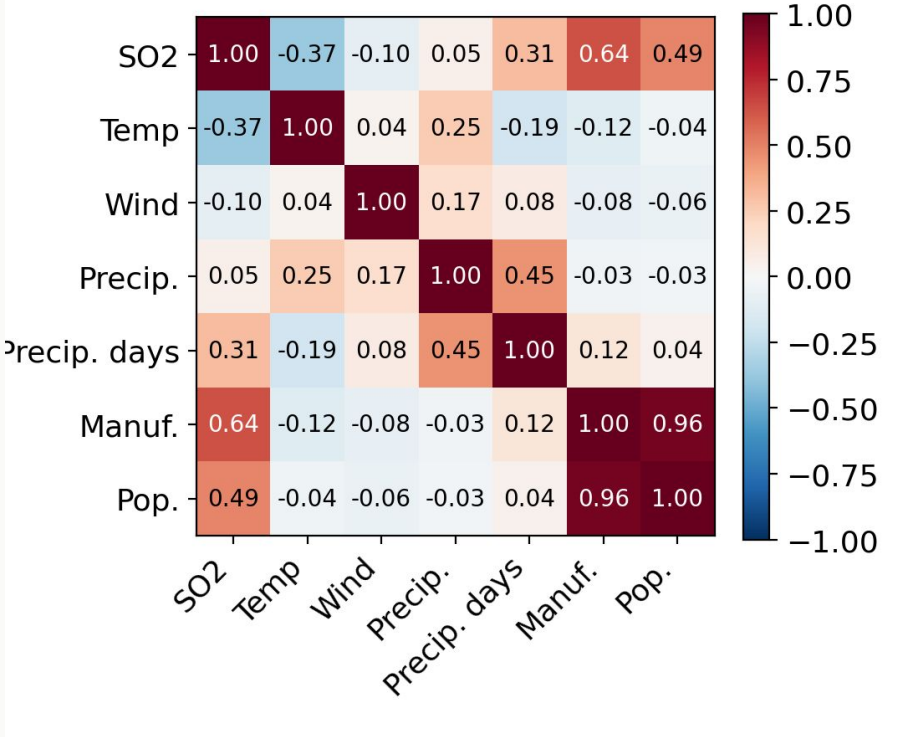
city-quarters (41 × 12)

9

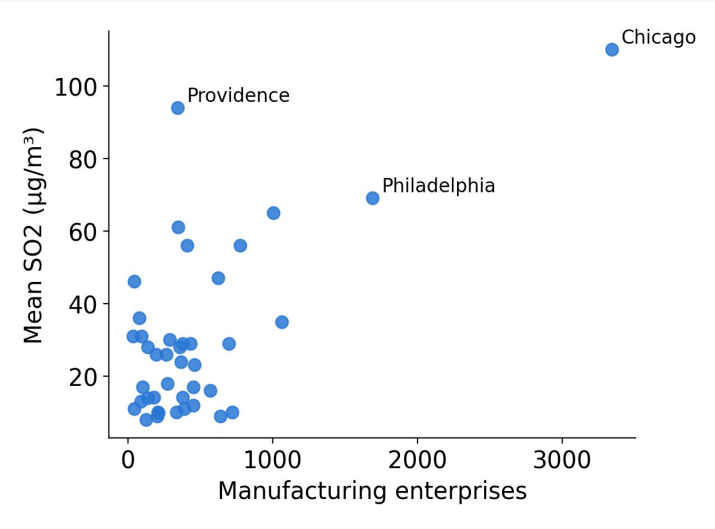
predictors incl. 3 season dummies

Not time-series forecasting: we predict SO2 from a city's conditions. Rows from one city are repeats, not independent.

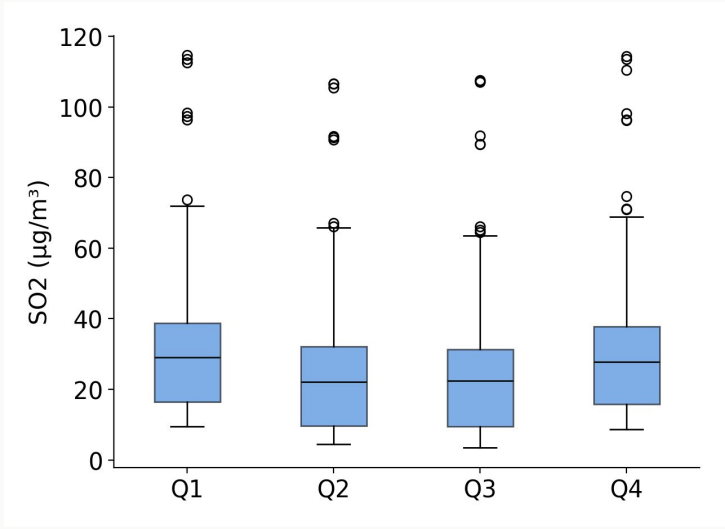
Exploring the relationships



Correlations: SO2 vs manufacturing 0.64, vs temperature −0.37



Bigger industrial cities have higher SO2

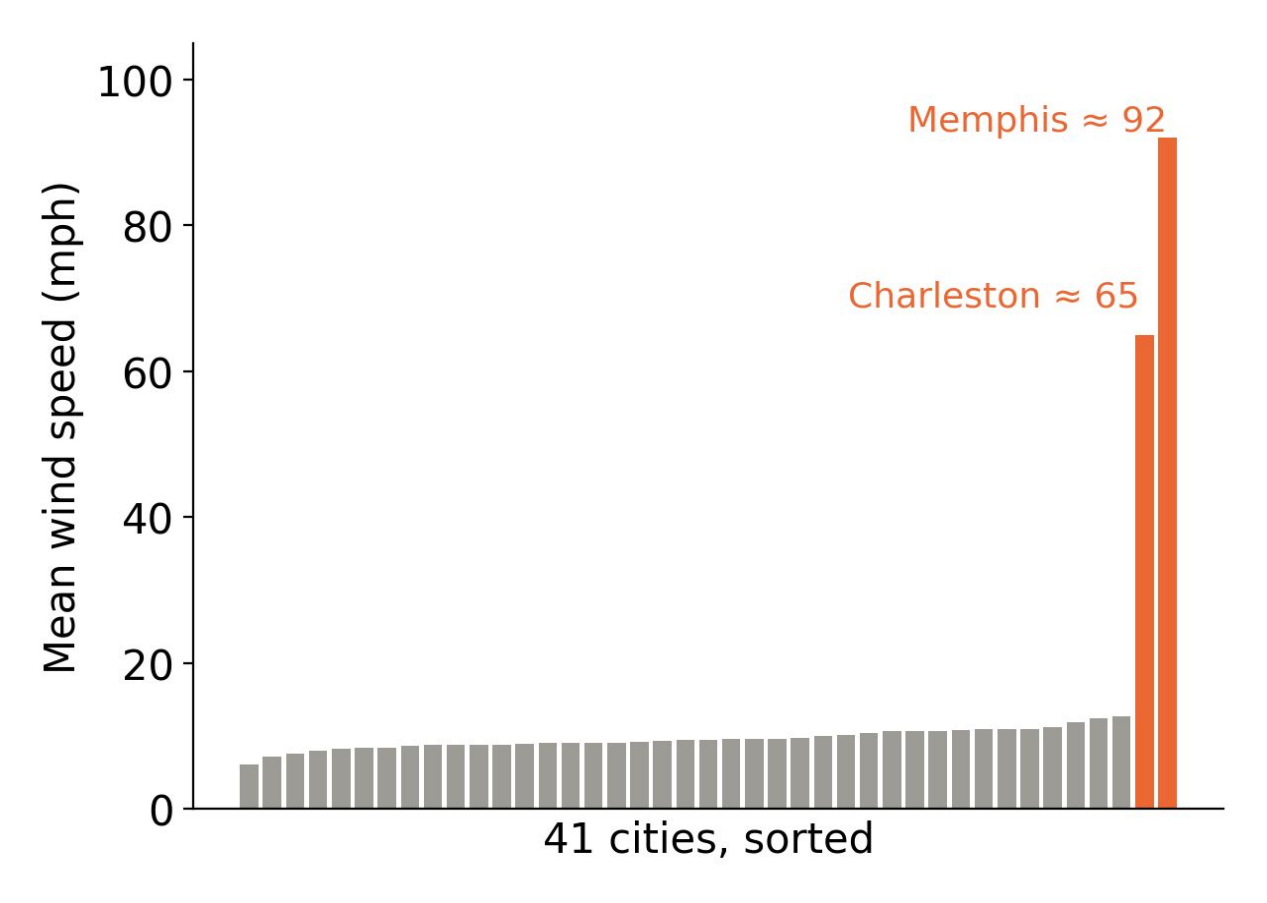


Season shifts SO2 only slightly

First look: industry and city size, then temperature. Manufacturing and population are 0.96 correlated: read them together.

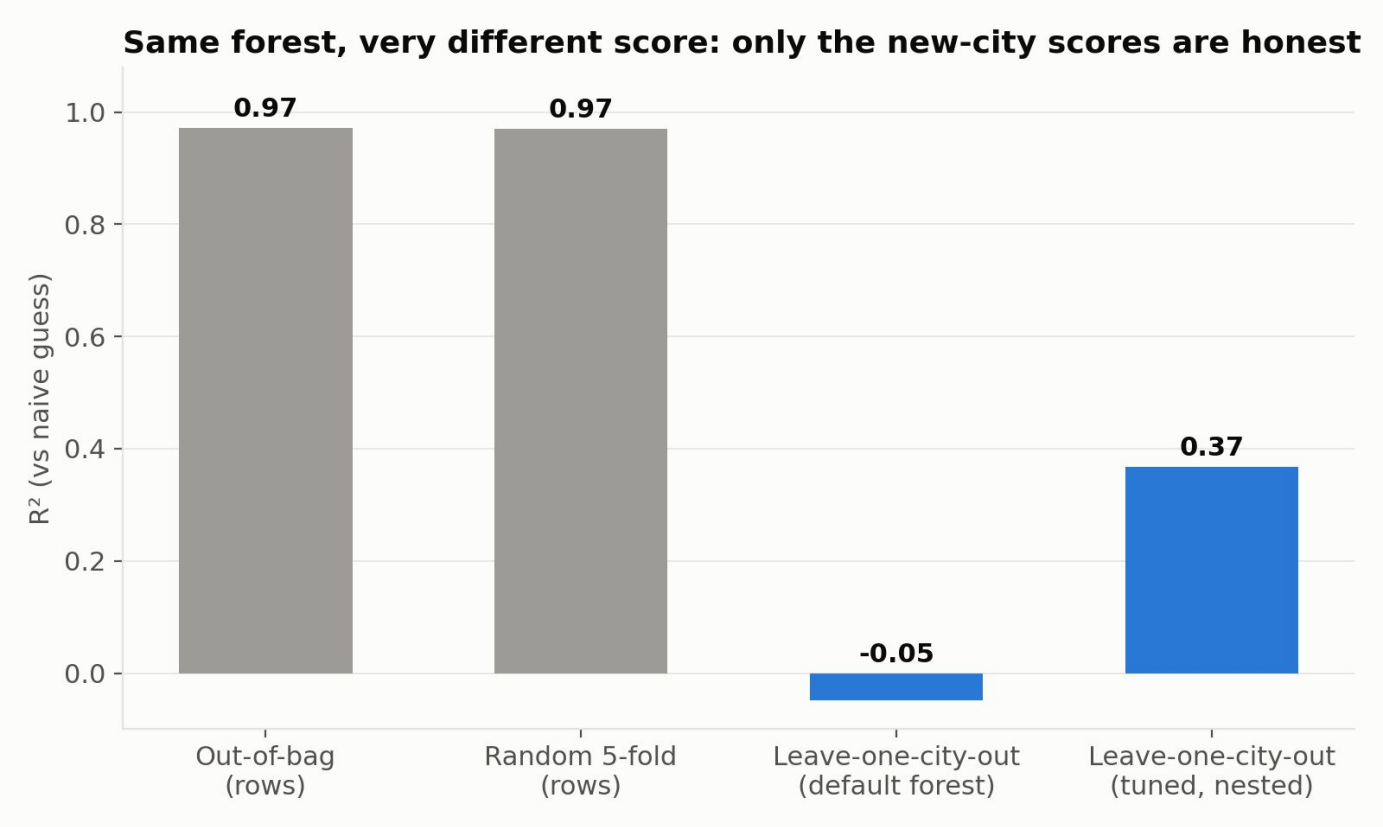
Answers: RQ1: exploring

Data quality: two impossible wind speeds



- Memphis ≈ 92 mph and Charleston ≈ 65 mph, in every quarter. Typical cities are 5–15 mph.
- Wind-speed coefficient: −0.08 with them, −2.6 without. Interpret with caution.
- Prediction error moves by ≤ 0.15 for regression, Ridge, Lasso and mixed models.

One honest test for every model



- Fit on 40 cities, predict the 41st, repeat 41 times.
- Tuning happens inside the 40 cities only.
- One shared scoring script for all four models.

How we score: k = 0, 1, 2

k = 0

A city with no SO2 readings. The hardest case, and our real question.

k = 1



We see the first quarter and shift every prediction by that miss.

k = 2



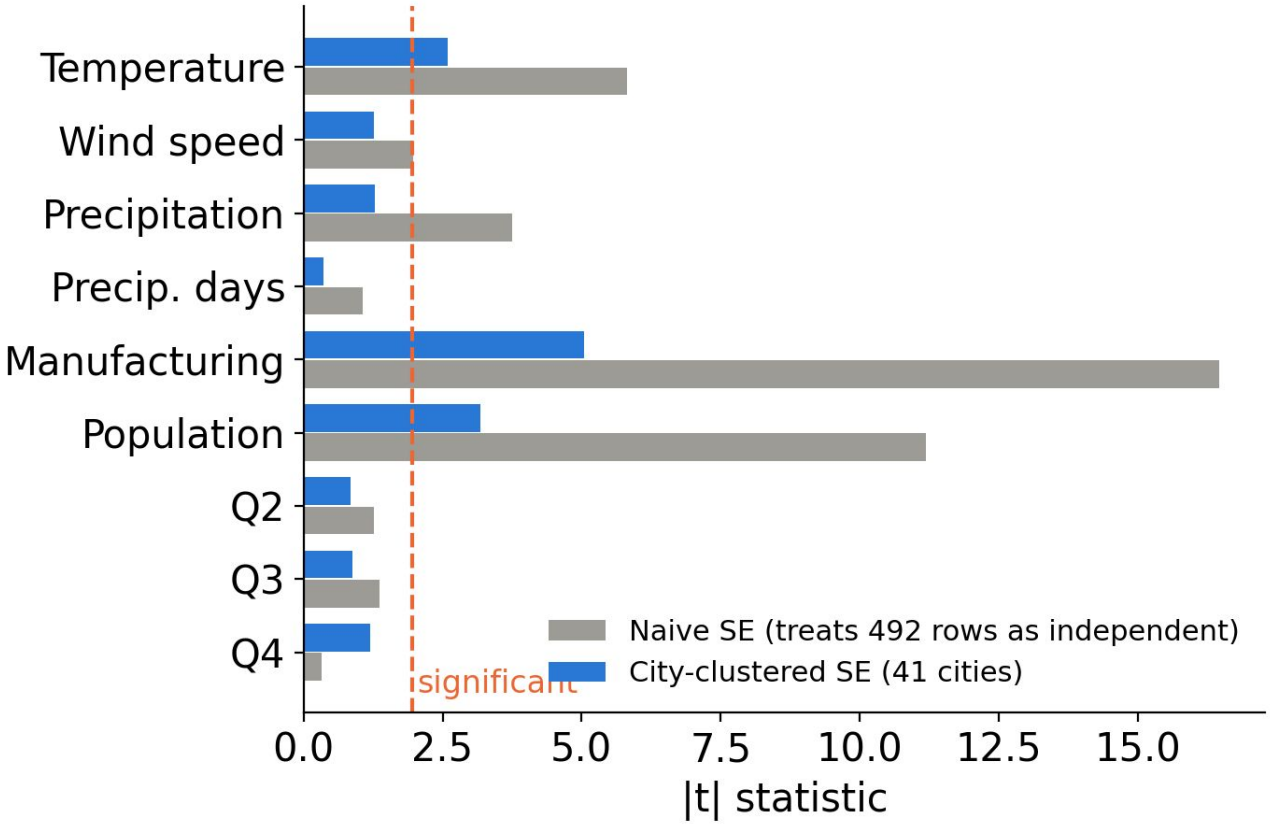
We see two quarters and shift by their average miss.

Dark = quarters we may see. Light = quarters we predict.

Benchmark: guess the average SO2 of the other 40 cities → RMSE 23.98. RMSE = typical size of an error in µg/m³ (SO2 averages 30).

Model 1: Multiple linear regression

Draws the best straight-line link between SO2 and all predictors at once.



- Clustering by city removes false significance for precipitation and wind speed.
- Temperature, manufacturing and population stay significant.
- Manufacturing and population: r = 0.96, read together.
- New city: RMSE 16.03, R^2 0.55.

Answers: RQ1 · RQ3

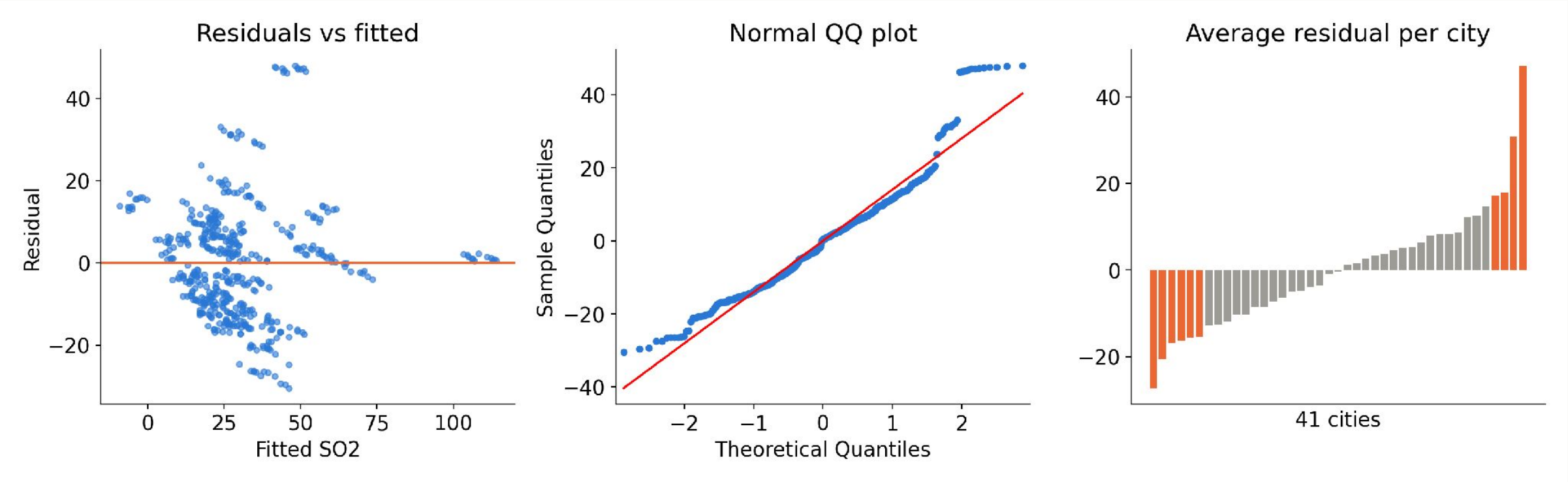
What the regression coefficients say

Predictor	Estimate	p (naive)	p (city-clustered)
Temperature (per °F)	−0.581	< 0.001	**0.010**
Wind speed (per mph)	−0.085	0.049	0.206
Precipitation	0.278	< 0.001	0.199
Precipitation days	0.123	0.288	0.714
Manufacturing enterprises	0.070	< 0.001	**< 0.001**
Population (thousands)	−0.046	< 0.001	**0.002**
Q2 vs Q1	3.09	0.207	0.401
Q3 vs Q1	3.45	0.172	0.382
Q4 vs Q1	0.58	0.750	0.234

- Orange = significant with city-clustered errors.
- Temperature, manufacturing and population survive; precipitation and wind speed do not.
- Manufacturing and population are 96% correlated: read them together.

Answers: RQ1 · RQ3

Checking the regression: the leftover error is city-shaped

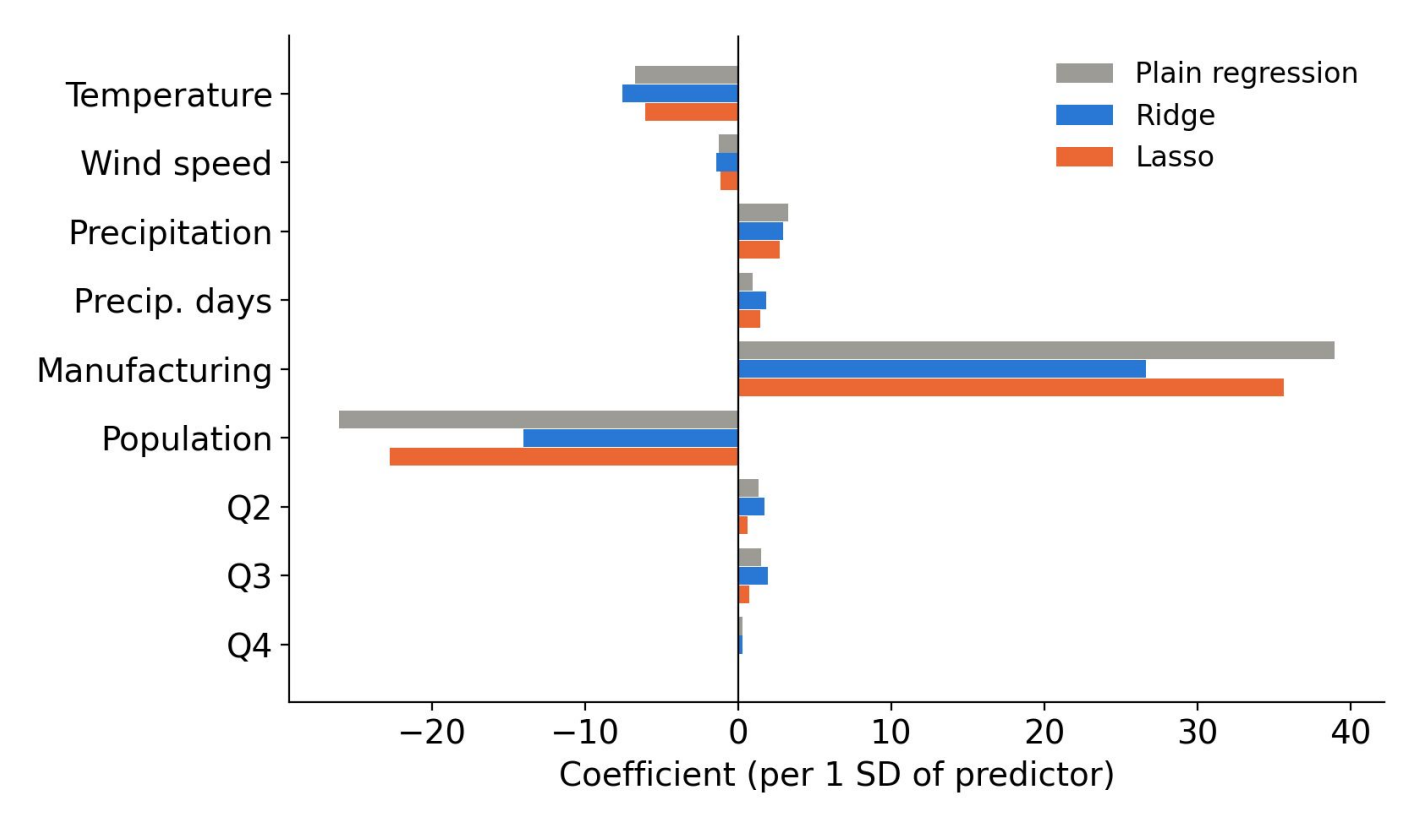


- Residuals are not normal and their spread is uneven (Shapiro and Breusch–Pagan tests, both $p < 0.001$).
- About 98% of the leftover error is a city-wide offset: a city is over- or under-predicted in all its quarters.
- This motivates the mixed-effects model, which gives each city its own baseline.

Answers: RQ1 · RQ3

Model 2: Ridge and Lasso

Regression with a penalty that shrinks coefficients to avoid overfitting; Lasso can set some to zero.

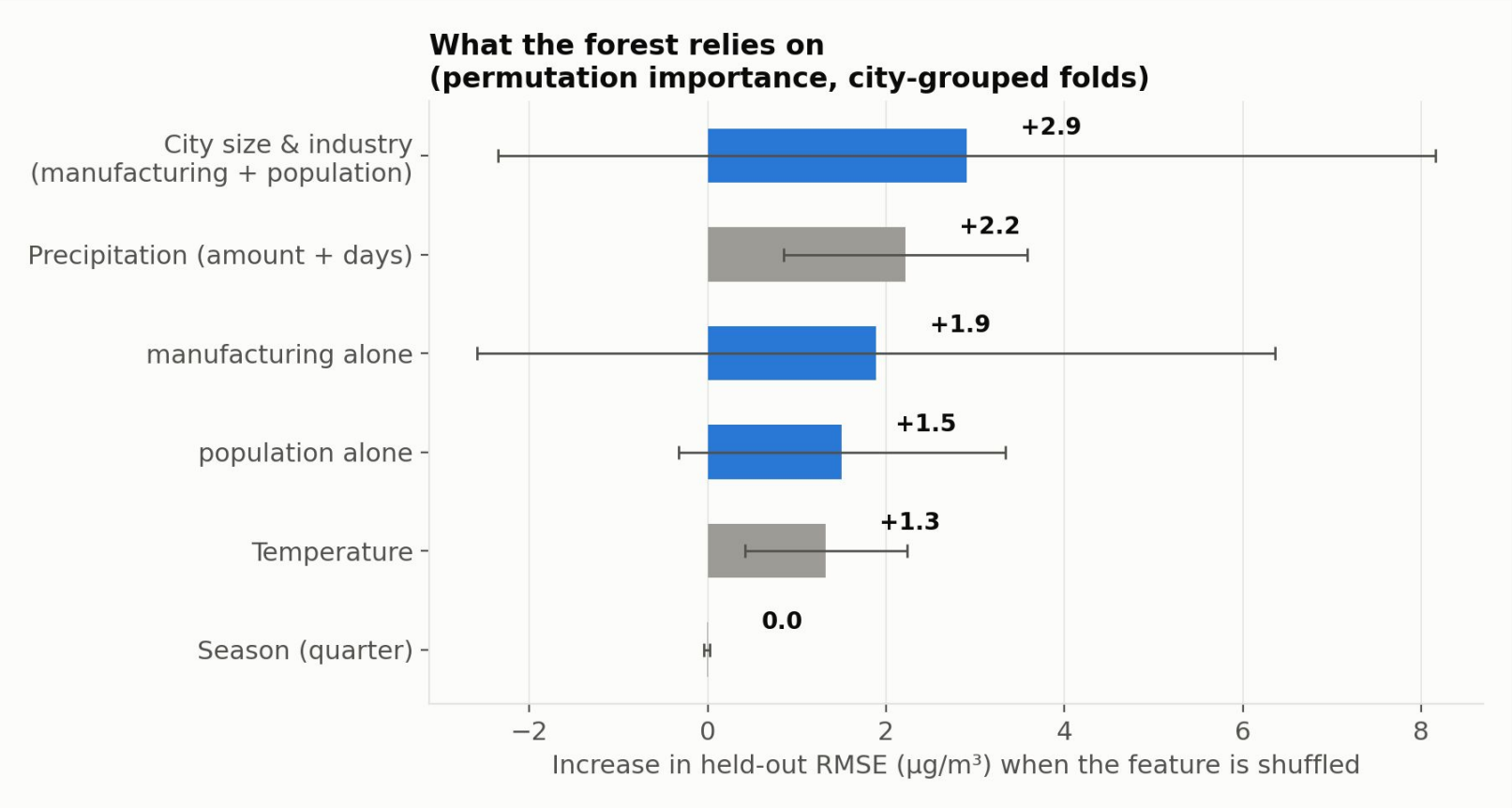


- Ridge pulls the correlated pair together: manufacturing 39 → 27, population −26 → −14.
- Lasso drops only the Q4 dummy and keeps the rest.
- New-city RMSE: Lasso 16.10, Ridge 16.39, plain regression 16.03: no gain from penalising.

Answers: RQ3

Model 3: Random forest

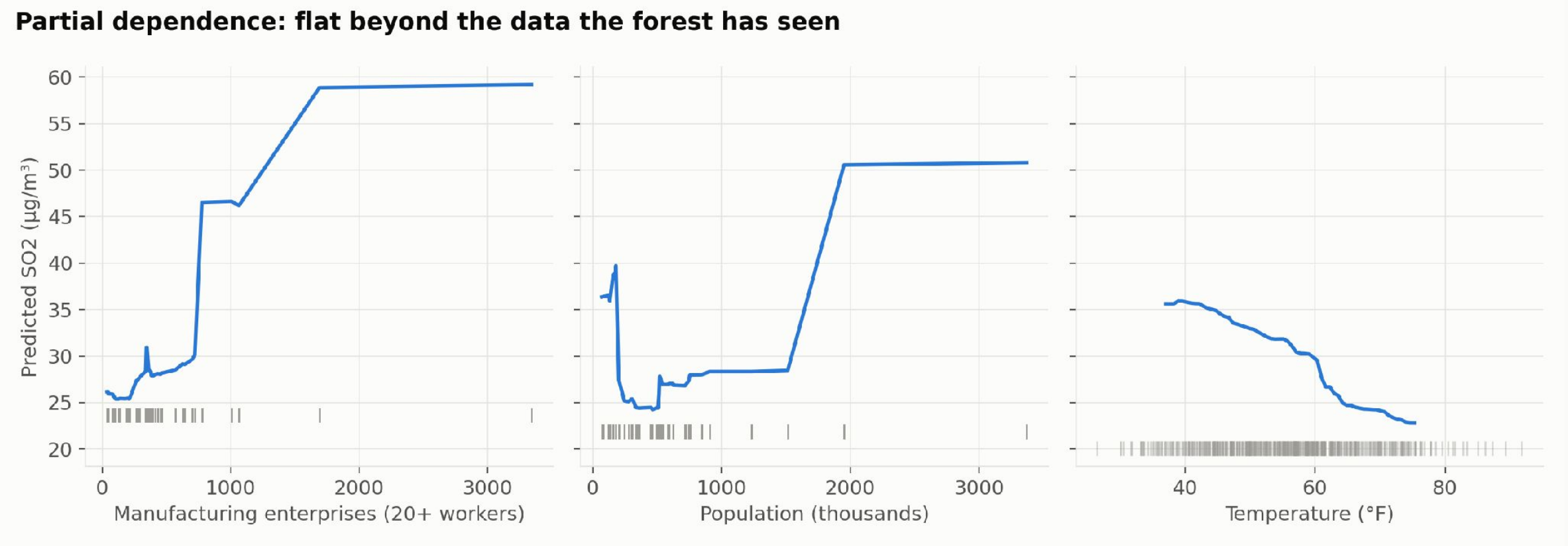
Hundreds of decision trees vote; averaging cancels out each tree's errors.



- City size and industry matter most; season adds nothing.
- Weakest model for a new city: RMSE 19.09.
- Trees cannot extrapolate: extreme cities like Chicago are under-predicted.

Answers: RQ3

Random forest: tuning and what it learned

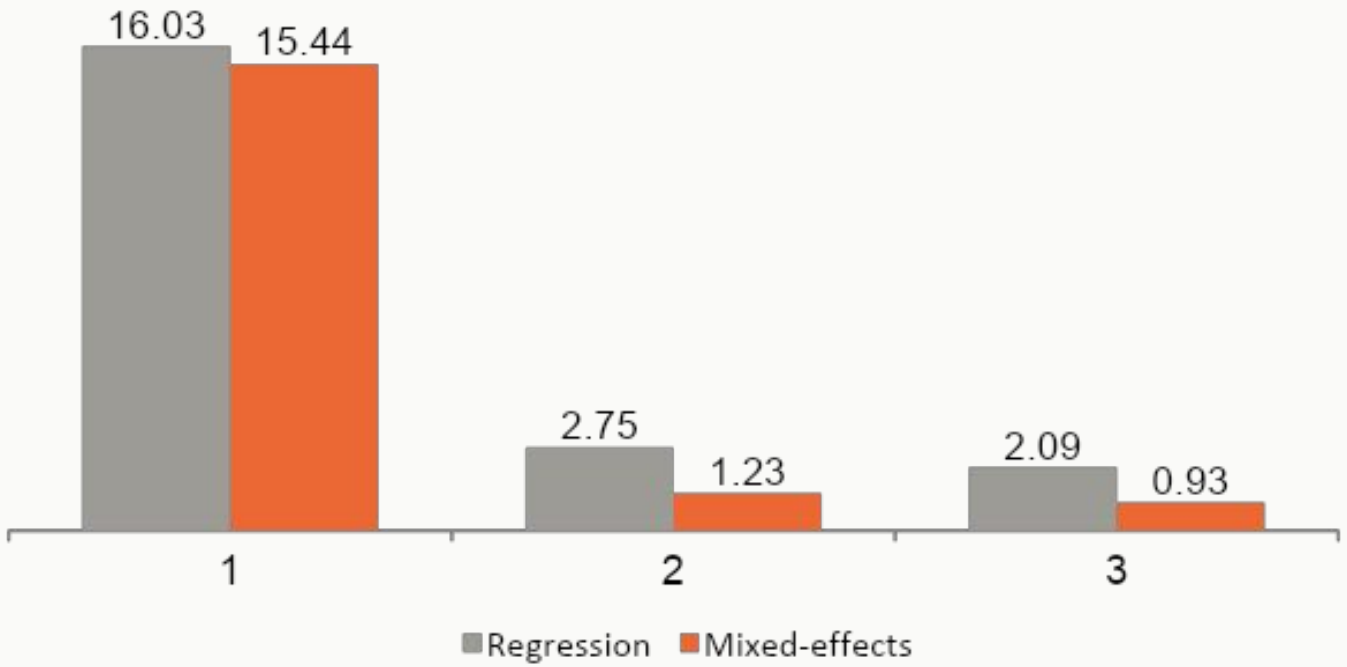


- Tuned with city-grouped CV inside the training cities: leaf size 10, 3 features per split (CV RMSE ≈ 19.3).
- Higher manufacturing and population → higher SO2; warmer → lower SO2. The curves go flat beyond the largest cities seen: trees cannot extrapolate.

Answers: RQ3

Model 4: Mixed-effects

Regression plus a personal baseline for each city.



Seeing even one quarter lets the model learn the city's own baseline: error falls from 2.75 to 1.23.

- Each city gets its own baseline (random intercept), fitted with REML.
- Almost all unexplained variation is between cities (ICC 0.997): one quarter pins down a city's level.
- Conformal band for a fully new city: ± 22.67 µg/m³.

Answers: RQ3

How to predict SO2 for a new city

1	**Collect the city's inputs**	Quarterly temperature, precipitation, wind, precipitation days; manufacturing count; population; which quarter.
2	**Plug into the fitted equation**	See below.
3	**Read the answer with its band**	Prediction ± 22.67 (90% band). One measured quarter? Add 0.997 × (measured − predicted) to every quarter.

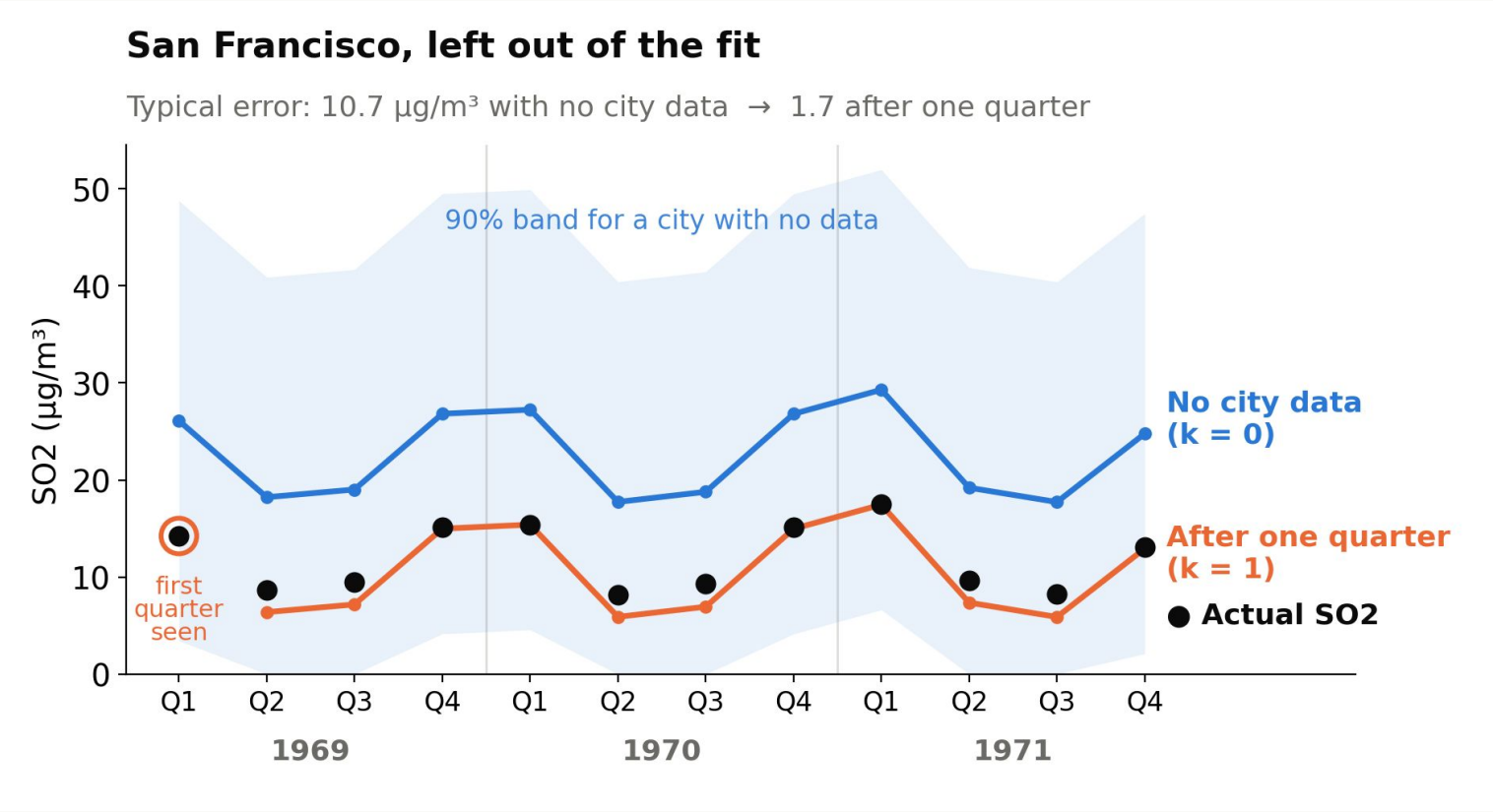
$$\hat{y} = 38.08 - 0.234\,T - 0.105\,W + 0.104\,R - 0.007\,D$$

$$+ 0.078\,M - 0.053\,P - 2.37\,Q_2 - 2.44\,Q_3 - 0.19\,Q_4$$

$\hat{y}$ = predicted SO2 (µg/m³) T temperature (°F) W wind speed (mph) R precipitation (inches) D precipitation days M manufacturing enterprises P population (thousands) Q_2, Q_3, Q_4 = 1 in that quarter, else 0 (Q1 is the baseline)

Answers: RQ2

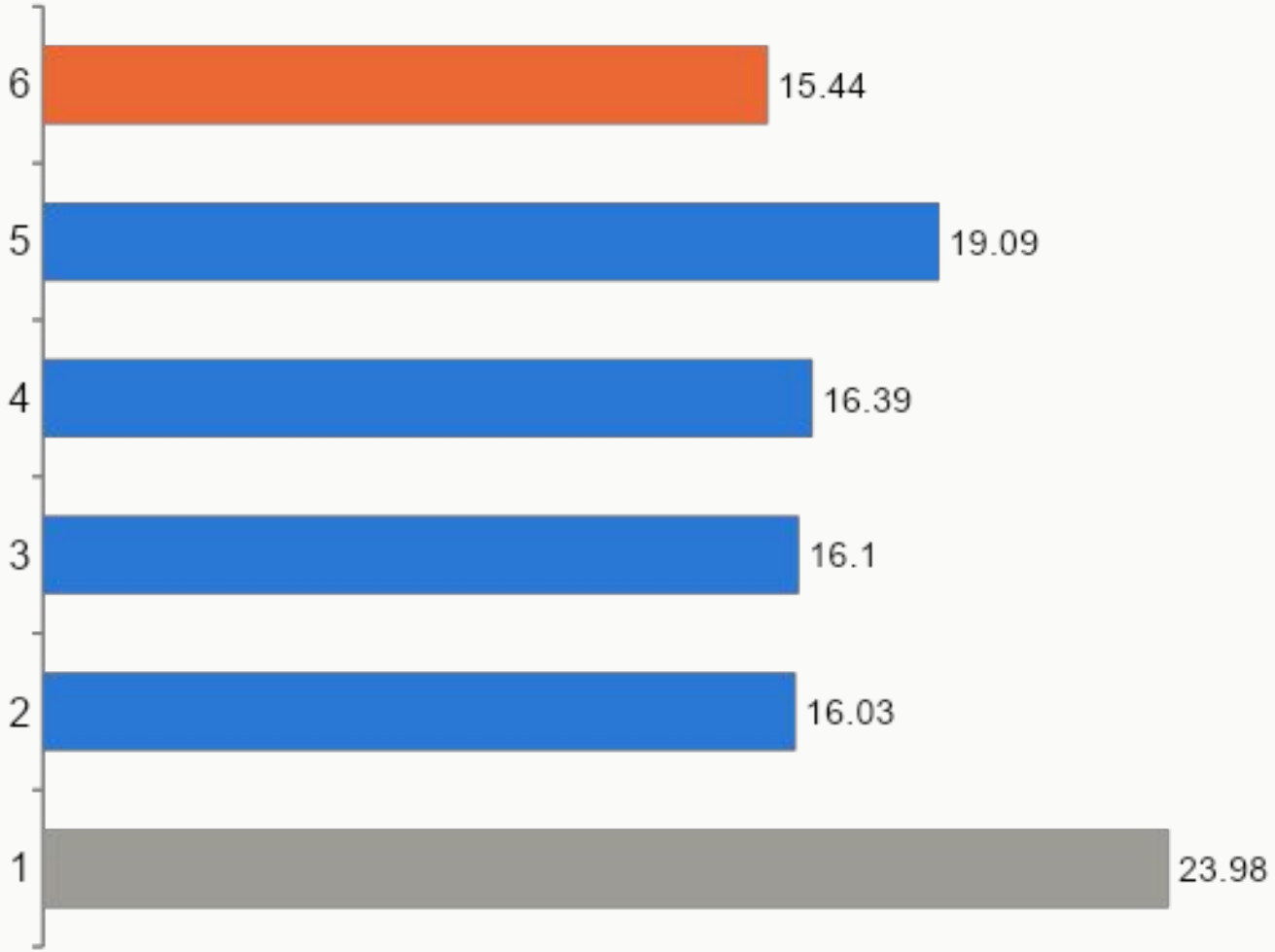
Worked example: San Francisco, left out of the fit



- San Francisco was left out of the fit: a typical city (error 10.7 = the median).
- No city data (blue): right seasonal shape, level about 10 too high; actuals inside the ±22.67 band in all 12 quarters.
- One quarter seen (orange): error falls from 10.7 to 1.7.

Answers: RQ2

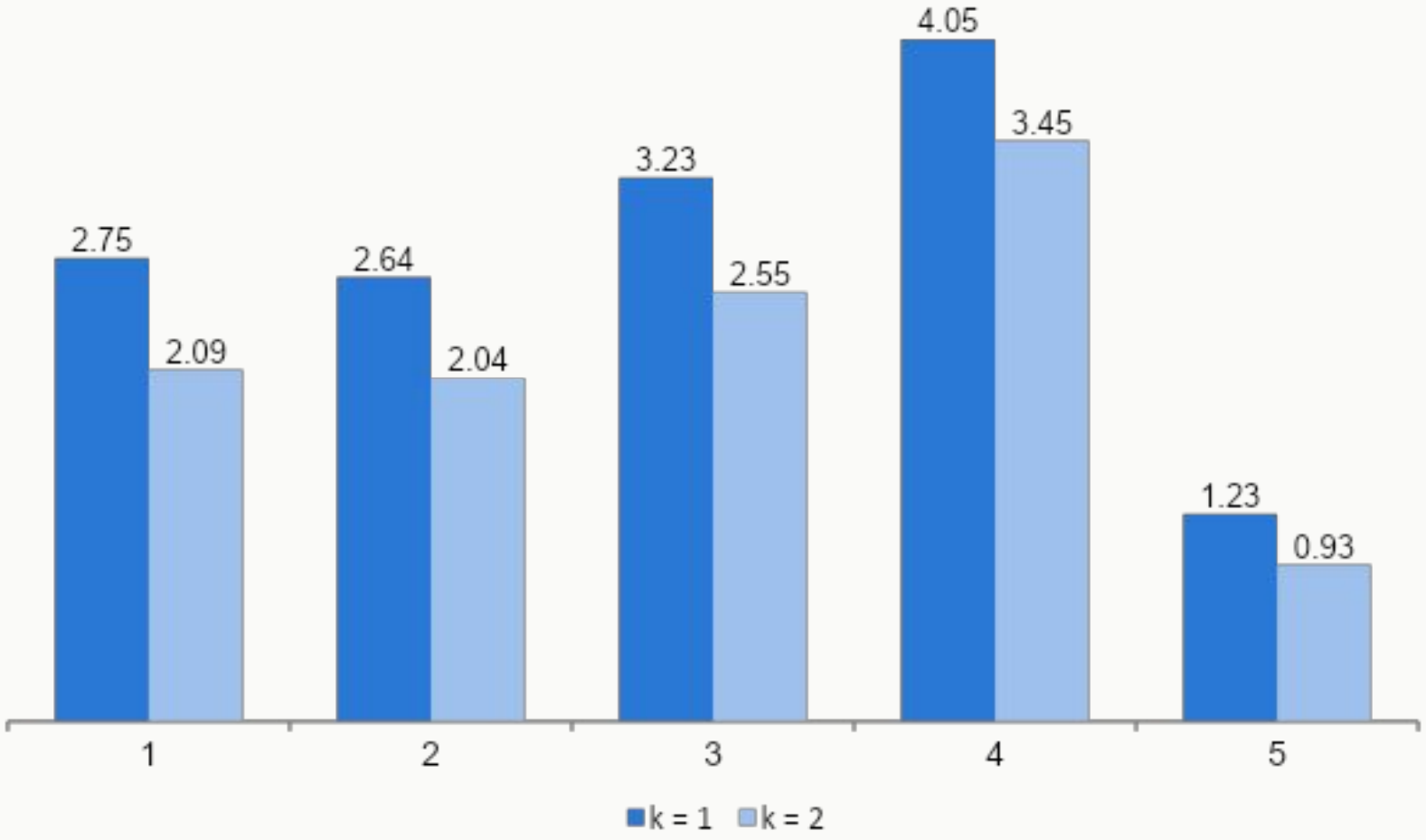
A brand-new city: the top four models are tied



- This tie is the finding: with no data from the city, model choice barely matters.
- 95% intervals: mixed 11.0–20.2, regression 11.7–20.6, Lasso 11.5–21.2, Ridge 11.6–21.8.
- Only the random forest (13.3–24.7) is clearly behind.

Answers: RQ2 · RQ3

Once we see one quarter, mixed-effects wins clearly



- Mixed-effects clearly best: its baseline estimate uses the revealed quarters properly.
- Random forest weakest at every k.
- One measurement is worth more than model choice.

Answers: RQ2 · RQ3

What drives SO2?

1	**City size and industry**	Largest effect in the regression and the forest.
2	**Temperature**	Colder quarters have higher SO2 (about −0.2 to −0.6 µg/m³ per °F, depending on the model).
3	**Precipitation**	Mixed evidence: not significant in clustered OLS, but positive in the mixed model and 2nd in the forest.
—	**Season, wind speed**	Small, and inconsistent between models (season has opposite signs).

Conclusion: did we answer our questions?

RQ1	Industry and city size drive SO2 first, then temperature. Season and wind speed add little.
RQ2	New city, no data: RMSE ≈ 15–16 (R^2 ≈ 0.55–0.59), about a third better than guessing: a rough screen. After one quarter: error ≈ 1.
RQ3	The top four models tie with no city data; mixed-effects wins clearly once a quarter is seen; random forest is weakest.
So what	Use the mixed-effects model; plain regression is the simple alternative.

Limitations and what we would do next

- Only 41 cities: new-city estimates stay uncertain (wide intervals).
- 1969–1971 data: the pollution era is different from today.
- Impossible wind speeds in two cities; some quarterly values are interpolated.
- Within a city, SO2 barely changes between quarters (ICC 0.997): most of what we predict is the city’s overall level.
- Next: more cities, corrected wind data, fuel-type predictors.